

An interface debonding law subject to viscous regularization for avoiding instability: Application to the delamination problems

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Abstract

The prediction of ply delamination in laminated composites is modeled by using interface elements. The numerical approach is based on the cohesive zone model, which is shown to provide an efficient description of the delamination growth. The theoretical study is performed in quasi-static regime and an implicit finite element scheme is used. A comprehensive 1D example shows that cohesive elements may induce numerical instability and that the use of a viscous regularization is relevant to suppress the instability and to obtain a reasonable numerical convergence.

This paper describes a technique that can be used to introduce damping into cohesive zone finite element simulations of crack nucleation and growth, with a view to avoiding convergence difficulties in quasi-static finite element simulations. A new kind of viscous regularization is proposed, which applies to the bilinear debonding model and has a limited rate dependency. An experimental study is performed by using DCB (double cantilever beam) samples for the mode I interlaminar fracture toughness test; a good agreement between numerical predictions and experimental results is obtained.

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1. Introduction

Polymer matrix composite materials are nowadays utilized in a large variety of structural applications in different industrial sectors such as space and aeronautics, naval constructions and civil engineering. Their well-known structural advantages lie in their high strength and stiffness-to-weight ratios compared to the usual structural metallic materials. Laminated composites are generally well suited for thin structures design and manufacture. However, service loads or initial manufacturing defects may gradually induce ply delamination, which, in turn, can lead to critical damages and a final dramatic drop of their mechanical properties. Several

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Nomenclature

Ψ	free Helmotz energy per volume unity
d	scalar damage variable
k_s and k_n	interface sliding and tension stiffnesses
Δ	displacement jump vector
Δ_i	the i th component of the displacement jump vector
Y_d	damage-deriving force
$\mathbf{t}$	interface traction vector
t_i	the i th component of the interface traction vector
$\mathcal{D}_e$	dissipated energy for a damaging process
δ	equivalent displacement jump
δ_0	onset equivalent displacement jump
δ_f	final equivalent displacement jump
Δ_i^0	pure mode i onset displacements jumps
$\Delta_i^0(\gamma)$	displacement jump associated to the onset of softening under mixed-mode loading, where $i = 1, 2$
$\Delta_i^f(\gamma)$	displacement jump associated to the complete interface degradation under mixed-mode loading
G_{ic}	critical energy release rate of mode i ($i = \text{I, II}$)
G_i	energy release rate for mode i in mixed-mode loading
G_c	critical energy release rate
$\tilde{t}$	equivalent scalar traction
d_v	viscous damage variable
$\Delta\tau$ and β	are parameters of viscosity
s	normalized time
ϕ	density of dissipated energy
ϕ_v	dissipated viscous energy
ϕ_v^{ideal}	ideal viscous energy

methods have been used to model and predict ply delamination growth and cracking. Griffith [1] introduced the linear elastic fracture mechanics (LEFM) to analyze the fracture behavior of isotropic materials. The extension of LEFM to orthotropic materials and composite materials started at the end of 1960s with the contribution of Sih and Liebowitz [2], for instance. The development of efficient numerical tools lead to the virtual crack closure technique (VCCT), initially proposed by Rybicki and Kanninen [3]. This type of analysis has been used to predict the delamination growth in laminated fiber-reinforced composites. However, the high computation cost, due to the finite element mesh refinement to model the crack initiation, is an important limitation of the method. The Cohesive Zone Model (CZM) was developed as a less computationally expensive alternative to the VCCT. CZM is a new approach to analyze the delamination process by finite elements and it can be developed within the framework of damage mechanics. The cohesive crack model was originally introduced by Dugdale and Barenblatt [4]. Hillborg et al. [5] developed the concept of tensile strength available for existing cracks to grow and/or initiate new cracks. Needleman [6] and Tvergaard [7] initiated the study of a crack growth in mixed modes, combinations of the crack opening modes are considered. Allix et al. contributed a series of papers dealing with crack growth along ply interfaces [8–10].

Finite element analysis (FEA) in quasi-static regime using CZM generally has to cope with convergence difficulties. For that, using an explicit FE code for the simulation of quasi-static delamination experiments can overcome to the problem of localization [11]. In implicit finite element code, and for certain mesh size and cohesive law parameters, numerical instabilities can occur during simulations. Various approaches can be utilized to solve these convergence problems. For general-purpose schemes, the Riks method [12] can be used to characterize the unstable branch of the solution during the snap-back. These schemes require some effort to implement and can give unphysical results in a real structural analysis. Other numerical approaches

can be found in the literature that eliminate localization (material instability) problems, based on the independence of the dissipated energy compared with mesh size. Indeed, there exist attempts to expand the local theory of continuum mechanics by the introduction of discontinuities of displacements or deformations [13]. Numerical implementation of this method requires finite elements and special formulation, close to the incompatible modes method. de Borst et al. [14] contributed uses these enhanced finite elements for the modeling of the damages in composite materials. A second method requires very important modifications within the framework of the standards continuum mechanics is the abandonment of the local theory in favor of the non-local theory [15]. However, numerical implementation of these models, proves much more complex and their application remains very limited.

Another numerical approach is the use of viscous regularization in order to overcome the localization problem, initially suggested by Needleman [16] for a plastic law. Chaboche et al. [17] and Gao et al. [18] have introduced a small, fictitious viscosity in the cohesive zone by using Tavergeard and Xu–Needleman's [19] interfacial constitutive law. This numerical tool is very effective for solving localization problems. However, this technique is particularly sensitive to the unrealistic effect of viscosity on the dissipation energy, hence on cohesive law.

In this paper, a cohesive and continuous viscous damage model is developed for the simulation of delamination in laminated composite. The goal is to limit the localization problems by introducing a term of viscosity into the cohesive law. Our main objective is to ensure that the dissipated energy remains independent of the mesh size. Hence, a minimum of dependence of the cohesive zone law to a viscosity parameter is guaranteed. The model presented here can be used with all debonding laws in order to completely solve the problem of localization. Furthermore, the model's implementation requires few modifications to a standard finite elements code.

We first review the basic aspects of the bilinear constitutive interface law model, which was developed in previous studies [20]. After that, we show how to introduce the viscosity term into the constitutive equations for cohesive interfaces. The viscosity parameter is added to the damage thermodynamics variables. Then, we point out the influence of the viscosity on both the cohesive zone behavior and convergence. Finally, we use the modified interface law to model a standard DCB specimen and prove the efficiency of the approach to solve the problem of the “solution jump” and instability.

2. Cohesive zone law

Interface zone laws are used to describe the relationships between tractions and displacement jumps through the interface during the debonding process. This section presents the bilinear cohesive law based on the studies of Geubelle and Baylor [21] and de Moura et al. [22]. We describe the bilinear traction, which links the normal (mode I) or shear (mode II) stresses, acting in the crack plane, and the respective normal or shear strains. The interface law was applied to model the ply delamination in laminated fiber-reinforced composites [23,24] and the behavior of adhesively bonded assemblies [25]. The cohesive model formulation is based on the damage mechanics of continuous media and LEFM.

2.1. Thermodynamics formulation

The cohesive law of the interface is classically expressed by traction–displacement jump (t – Δ) relationship written in the local $(\vec{e}_s, \vec{e}_n)$ basis. The thermodynamic potential in 2D is as follows [20]:

$$\Psi(\Delta, d) = \frac{1}{2}(1-d) \left(k_s \Delta_1^2 + k_n \langle \Delta_2 \rangle_+^2 \right) + \frac{1}{2} k_n \langle \Delta_2 \rangle_-^2 \quad (1)$$

where Ψ is free Helmholtz energy per unit volume, $d \in [0, 1]$ denotes a scalar damage variable, symbols $\langle \cdot \rangle_\pm$ stand for the positive and negative part of the argument defined as: $\langle x \rangle_\pm = (x \pm |x|)/2$, the scalar parameters k_s and k_n are a penalty stiffnesses in shear and traction, respectively, and Δ_i stands for i th component of the displacement jump vector, Fig. 1. The constitutive equations of thermodynamic forces for the model are obtained by differentiating the free energy with respect to the state variables, that is

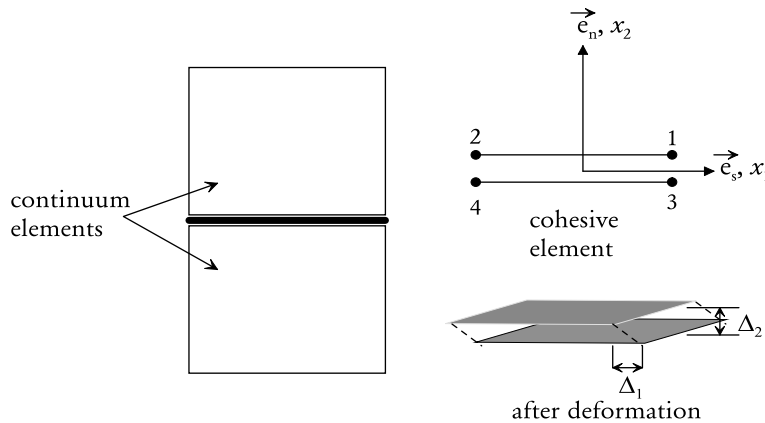


Fig. 1. Interfacial representation.

$$\begin{cases} t_i = \frac{\partial \psi}{\partial \Delta_i}, \quad i = 1, 2 \Rightarrow t_1 = (1-d)k_s \Delta_1; & t_2 = (1-d)k_n \langle \Delta_2 \rangle_+ + k_n \langle \Delta_2 \rangle_- \\ Y_d = -\frac{\partial \psi}{\partial d} = \frac{1}{2}(k_s \Delta_1^2 + k_n \langle \Delta_2 \rangle_+^2) \end{cases} \quad (2)$$

where Y_d is the elastic work for a displacement jump Δ , t_i represents the i th component of the stress vector. The expression of the dissipated energy for a damaging process is given by

$$\mathcal{D}_e = Y_d \dot{d} \quad (3)$$

where $\dot{d}$ is the damage rate. Finally, the classical Clausius–Duhem inequality states that the intrinsic dissipation due to the debonding process must remain semi-positive. That is, $\mathcal{D}_e \geq 0$. In order to consider the coupling between the different modes we define a norm of the displacement jump vector called the equivalent displacement jump. It can be used to compare different levels of the displacement jump and employed to define concepts such as “loading”, “unloading” and “reloading”. The equivalent displacement jump is a non-negative and continuous function defined as (Mazars [26])

$$\delta = \sqrt{\alpha^2 \Delta_1^2 + \langle \Delta_2 \rangle_+^2}; \quad \text{with } \alpha^2 = \frac{k_s}{k_n} \quad (4)$$

The Clausius–Duhem inequality imposes the following restrictions on the damage variable d : $\dot{d} \geq 0$, which ensures the irreversibility of damage. The growth of the damage is described by an increasing scalar function defined by Turon et al. [24]

$$d(\delta) = \begin{cases} 0 & \text{for } \delta < \delta_0 \\ \frac{\delta_f(\delta - \delta_0)}{\delta_f - \delta_0} & \text{for } \delta_0 \leq \delta < \delta_f \\ 1 & \text{for } \delta \geq \delta_f \end{cases} \quad (5)$$

where δ_0 is the onset displacement jump. It represents the initial value of the displacement jump when the damage starts. The initial damage threshold is derived from the formulation of the initial damage surface or criterion. δ_f is the final displacement jump. The material is considered as completely damaged when δ_f is reached. It is given by the model of the propagation surface or criterion.

2.2. Criterion of the onset of delamination

When the interface is subject to pure I, or II loading modes, the delamination occurs if the interlaminar associated stress reaches its maximum interfacial value. Obviously, under mixed-mode loading, delamination onset may occur before those maximum values are attained. The interaction between the stress components under mixed-mode loading has to be taken into account by using a multi-axial stress criterion given by

$$\left(\frac{t_1}{t_1^0}\right)^2 + \left(\frac{\langle t_2 \rangle_+}{t_2^0}\right)^2 = 1 \quad (6)$$

where t_2^0 and t_1^0 are maximum interlaminar stress vector components along x_2 and x_1 directions, respectively. Turon et al. [24] proposed the introduction of a new variable defined as follows to describe the interaction process

$$\gamma = \frac{A_1}{A_1 + \langle A_2 \rangle_+} \quad (7)$$

By using Eqs. (2), (4), (5), (6) and (7) one can determine the initial values of damage in mixed-mode loading

$$\begin{cases} \delta_0 = A_1^0 A_2^0 \sqrt{\frac{(1-\gamma)^2 + (\alpha\gamma)^2}{((1-\gamma)A_1^0)^2 + (\gamma A_2^0)^2}} \\ A_1^0(\gamma) = A_1^0 A_2^0 \frac{\gamma}{\sqrt{((1-\gamma)A_1^0)^2 + (\gamma A_2^0)^2}} \\ A_2^0(\gamma) = A_1^0 A_2^0 \frac{(1-\gamma)}{\sqrt{((1-\gamma)A_1^0)^2 + (\gamma A_2^0)^2}} \end{cases} \quad (8)$$

where A_i^0 is the pure mode's i th onset displacements jumps, $A_i^0(\gamma)$ is displacement jump associated with the onset of softening under mixed-mode loading, where $i = 1, 2$.

2.3. Propagation criterion

Delamination propagates when the energy release rate attains its critical value under pure mode I or mode II fracture. Generally, delamination growth occurs under mixed-mode loading. In that case, the delamination process may take place before any of the energy release rate components reaches its individual critical value. The most widely used criterion to predict propagation of delamination under mixed-mode loading is the power law criterion proposed by Reeder [27]

$$\left(\frac{G_I}{G_{Ic}}\right)^n + \left(\frac{G_{II}}{G_{IIc}}\right)^n = 1 \quad (9)$$

where n is an empirical parameter which is considered as characteristic of the existing coupling phenomenon between the modes; G_{ic} is the critical energy release rate of mode i ($i = I, II$). The LEFM allows us to build up the relationship between the interface behavior and the total energy release rate in mixed-mode loading G , which is simply defined as the area under the equivalent scalar traction versus the equivalent displacement jump curve $(\tilde{t}-\delta)$, that is

$$G = \int_0^{\delta(t)} \tilde{t} d\delta \quad (10)$$

where $\tilde{t} = \sqrt{\alpha^{-2}t_1^2 + t_2^2}$ is the equivalent scalar traction, $\delta(t)$ is the equivalent displacement jump at time t . The critical energy release rate is defined by the final equivalent displacement δ_f . In the bilinear softening law, G_c is as follows:

$$G_c = \int_0^{\delta_f} \tilde{t} d\delta = G_I + G_{II} = \frac{1}{2} k_n \delta_0 \delta_f \quad (11)$$

G_i is the energy release rate for mode i in mixed-mode loading. For a given mixed-mode γ , we obtain for each mode

$$G_I = \frac{1}{2} k_n A_2^0(\gamma) A_2^f(\gamma); \quad G_{II} = \frac{1}{2} \alpha^2 k_n A_1^0(\gamma) A_1^f(\gamma) \quad (12)$$

where $A_i^f(\gamma)$ is the displacement jump associated to the complete interface degradation under mixed-mode loading. From Eqs. (4), (5), (9), (11) and (12) the energy release rate in mixed-mode loading can be derived as

$$G_c = G_{Ic} G_{IIc} \frac{(\alpha\gamma)^2 + (1-\gamma)^2}{[(G_{Ic}(\alpha\gamma)^2)^n + (G_{IIc}(1-\gamma)^2)^n]^{1/n}} \quad (13)$$

From Eq. (11) the expression of the equivalent displacement jump, corresponding to the total ply debonding under mixed-mode loading δ_f , can be retrieved as

$$\delta_f = \frac{2G_c}{k_n \delta_0} \quad (14)$$

By using Eqs. (7), (12) and (13), the total debonding displacement jump can be written as

$$\begin{cases} \Delta_1^f(\gamma) = \frac{\delta_f \gamma}{\sqrt{(1-\gamma)^2 + (\alpha\gamma)^2}} \\ \Delta_2^f(\gamma) = \frac{\delta_f (1-\gamma)}{\sqrt{(1-\gamma)^2 + (\alpha\gamma)^2}} \end{cases} \quad (15)$$

A quadratic solid plane element with four nodes and two Gaussian points has been selected for the numerical model. The implementation of the bilinear behavior has been carried out as user element subroutine (UEL) in ABAQUS Standard v6.5 [29]. Computations have been performed using a various step sizes and the Newton–Raphson algorithm.

3. Instability of 1D cohesive model

The instability due to the constitutive cohesive law (t – Δ) is a real problem for the implicit finite elements code. The use of the Newton–Raphson method is not adequate to solve problems with unstable branches of solutions. This instability does not only affect the uniqueness of the FEM solution but it also creates a discontinuity of the solution which we call “solution jump”. To clarify this particular point, this section is devoted to the material instability problem. A comprehensive mechanical example will be utilized to explain the origin of snap-back instability when the bilinear softening law is used.

3.1. Material instability problem

Let us consider the particular case of two elastic trusses with Young modulus E , length a and unit width (Fig. 2). The trusses are connected together by a cohesive element. It is assumed that the two material blocks are stretched along the loading x -axis only (imposed displacement $\bar{u} \geq 0$). Our goal here is to solve this mechanical problem analytically, i.e. to find the exact displacement jump solution. From the equilibrium equations of the system, one can derive the cohesive traction t as

$$t = \frac{E(\bar{u} - \Delta)}{2a} \quad (16)$$

Let Δ the displacement jump. At the interface, the behavior law is given by a bilinear debonding model (t – Δ)

$$\frac{t}{k\Delta_0} = \frac{t}{t_0} = \begin{cases} \frac{\Delta}{\Delta_0} & \text{if } \frac{\Delta}{\Delta_0} < 1 \\ \frac{\frac{\Delta_f}{\Delta_0} - \frac{\Delta}{\Delta_0}}{\frac{\Delta_f}{\Delta_0} - 1} & \text{if } 1 \leq \frac{\Delta}{\Delta_0} < \frac{\Delta_f}{\Delta_0} \\ 0 & \text{if } \frac{\Delta}{\Delta_0} \geq \frac{\Delta_f}{\Delta_0} \end{cases} \quad (17)$$

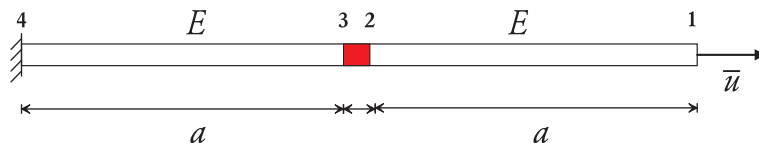


Fig. 2. Truss with cohesive zone subjected to a traction.

where Δ_0 and Δ_f represent the onset and final displacement jump in the cohesive zone, respectively. These parameters are defined by the onset criterion and the delamination propagation. Thus the complete set of equations is composed of Eqs. (16) and (17) and the variable is the displacement jump Δ . Finally, the analytical solution can be expressed as a function of the imposed displacement $\bar{u}$ and a normalized constant Λ , such as

$$\Lambda = \frac{E\Delta_0}{2at_0}; \quad \Lambda_c = \frac{1}{\alpha - 1} \quad \text{and} \quad \alpha = \frac{\Delta_f}{\Delta_0} \quad (18)$$

Fig. 3a shows the dependence between the normalized stress and the imposed displacement. Fig. 3b presents the normalized displacement jump versus the imposed displacement for several values of Λ . First, we notice that for $\Lambda \geq \Lambda_c$, the local response and the traction at the interface are always stable; however when $\Lambda < \Lambda_c$, as $\bar{u}$ increases we can predict that a displacement jump or tension will occur. It appears that the mechanical system is becoming unstable. In fact, this “solution jump” does not lead to a loss of uniqueness (Chaboche et al. [17]) but just to a sudden jump of the displacement and a drop of the stress. In this case, the interface releases the elastic energy stored in the bulk material rapidly, involving the instantaneous

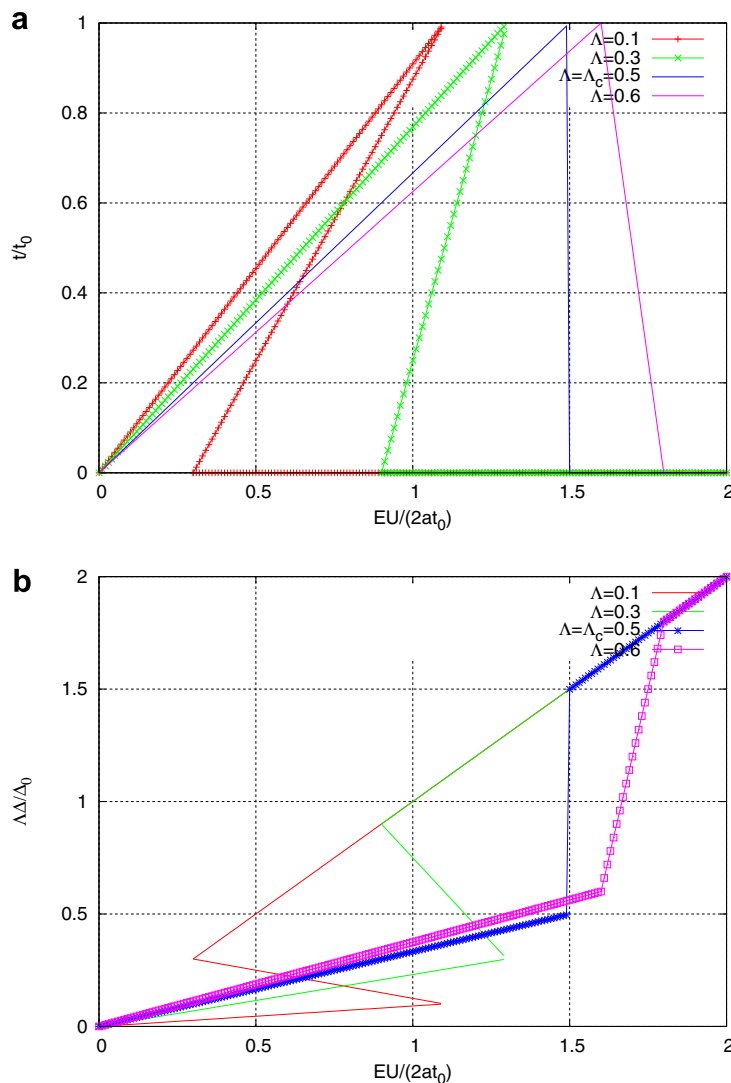


Fig. 3. Solution of the 1D problem of a truss in tension: (a) traction in the cohesive zone and (b) displacement jump.

breaking of the cohesive zone. The material instability of the cohesive model depends on the elastic parameters, the length of the bars (mesh dependency) and the constitutive parameters of the softening bilinear law. It is now possible to set up an instability criterion of this system

$$\text{The system is unstable if : } A < A_c \iff \frac{at_0^2}{E} > \phi - \frac{1}{2}k\Delta_0^2 \equiv \phi_d \quad (19)$$

where $\phi = \frac{1}{2}k\Delta_0 A_f$ is the area under the curve $(t-\Delta)$ interface law, which defines the critical energy release rate G_c , the ratio at_0^2/E is the elastic strain energy per unit of surface stored in the two bars. The criterion of instability shown in Eq. (19) postulates that: if the elastic energy stored in the elastic bars is higher than the effective cohesive energy ϕ_d , then the system is unconditionally unstable. This leads to a convergence problem in the finite element analysis.

3.2. Solving the instability problem

Several authors proposed different approaches to suppress the “solution jump”. Possible methods are listed in Chaboche et al. [17]. Allix [28] suggested to introduce a new concept of damage rate called “delay effect”. However, Chaboche notes, in the case of Allix’s model, that the rate dependency is very strong, at least in the high-rate regime. Chaboche introduced interface models which are based on viscous parameters to suppress the occurrence of the solution jumps. This approach induces a strong loading rate dependency into the structural analysis. He noted also that the law, previously proposed by Allix for the damage, induces a very sensitive dissipated viscous energy. by increasing the rate loading, Allix’s model converges but with an important dissipation. Gao and Bower [18] introduced an additional viscous term in a stress component as additional viscous traction so that the irreversibility dissipation condition is no more fulfilled. Our approach here aims to regularize the instability of the system by adding a viscosity to the cohesive model. In the present model, we propose an expression of the damage rate which allows the regularization for any rate. This artificial dissipation introduces a mechanism to absorb the excess of material bulk energy. In other words, it makes it possible to increase the effective energy ϕ_d ; thereby Eq. (19), which is the origin of the instability, is no more satisfied.

3.2.1. Damage model with viscosity

To suppress the “solution jump”, the bilinear softening law can be affected by the viscous parameter. In the present model, the damage variable d is called d_v and is displacement jump rates dependent.

The new damage variable will be expressed in mixed-mode loading as

$$d_v = d - (d_I + d_{II}) \quad (20)$$

where d is a continuous function defined by Eq. (5) in the case of the bilinear softening law, d_I and d_{II} stand for the viscous damage variables for mode I and II, respectively. This new formulation of the damage variable allows the increase of the density of dissipated energy in other words, the energy of cohesive model. These parameters are defined by

$$d_{I,II} = \Delta\tau_{I,II} \frac{\dot{\Delta}_{2,1}}{\Delta_{2,1}} \quad (21)$$

where $\Delta\tau_{I,II}$ are parameters of viscosity related to the stress by: $\frac{\sigma}{t_0} = \Delta\tau \frac{\dot{\Delta}}{\Delta_0}$. This relationship is similar to a law of a viscous fluid, $\sigma = \eta \dot{\epsilon}$, with η identified as $\Delta\tau$ and $\dot{\Delta}_{2,1}$ are the displacement jump rates of the interface in mode I and II, respectively. Injecting Eq. (21) to the cohesive law ensures the stability of the system of Fig. 2 for a minimum of dissipated energy (subject treated below). The selection of $\Delta\tau_{I,II}$ values is critical since it will control the behavior of the system in terms of loading rate. The thermodynamic admissibility of the modified damage variable fulfils the Clausius–Duhem condition, $\dot{d}_v \geq 0$. Therefore, the intrinsic dissipation of the debonding process remains positive. For physical reasons, one can set $d_v \in [0, 1]$; that is, the interface is undamaged or completely damaged when $d_v = 0$ or $d_v = 1$, respectively. From Eq. (2) one can obtain the bilinear traction–displacement jump relationship in mixed-mode loading

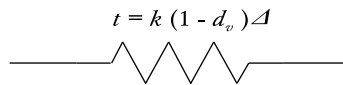


Fig. 4. Diagram of the cohesive model with viscoelastic behavior.

$$\begin{cases} t_1 = (1 - d_v)k_s \Delta_1 \\ t_2 = (1 - d_v)k_n \langle \Delta_2 \rangle_+ + k_n \langle \Delta_2 \rangle_- \end{cases} \quad (22)$$

To investigate the influence of the viscosity in detail, we perform a refined analysis of the 1D problem of (Fig. 2). Fig. 4 shows the rheological diagram of the 1D cohesive model where the bilinear softening law is affected by a new damage variable. Our main objective is to prove that finite element analysis has no difficulty to converge when a viscous term is added to the constitutive bilinear law.

It is convenient to assume that the displacement occurring at the end of the truss is a linear function of time, i.e. $\bar{u} = V_0 \tau$, where τ denotes the time and V_0 is the loading rate. From Eqs. (20)–(22) the traction can be related to the interface separation for 1D problem by

$$t = (1 - d_v)k\Delta, \quad \text{where } d_v = d - \Delta\tau \frac{\dot{\Delta}}{\Delta} \quad (23)$$

where $\Delta\tau$ is a parameter of viscosity, d is the scalar damage variable. We consider here the following normalized variables:

$$s = \frac{\tau}{\Delta\tau}; \quad \beta = \frac{V_0}{\Delta_0} \Delta\tau \quad (24)$$

where s and β represents the normalized time and viscous parameter, respectively.

Fig. 5 shows the response of cohesive zone for different β values. These results are obtained for $\Delta < \Delta_c$ (Eq. (19)). It can be seen that the system can be regarded as unconditionally unstable. Fig. 5a shows a displacement jump Δ , versus normalized imposed displacement, $\bar{u}$, using both finite element and analytical analysis. For the particular case of $\beta = 0$ (no viscosity) the numerical solution of the problem, i.e. a truss subjected to a continuous displacement, cannot be determined by a Newton–Raphson algorithm because, Eq. (19) is always true. However, the introduction of viscosity into the cohesive model modifies the response of the interface to the level of “displacement jump” and permits to solve this instability, although the time step is small. Fig. 5b shows the effect of viscosity on the cohesive zone behavior (traction–separation law). It is noted that increasing viscosity increases the interface strength.

In order to analyze the effect of the regularization term on the damage variable, Fig. 6a presents the new viscous damage evolution versus the imposed displacement for different values of the viscosity. We notice that for small values of the viscous parameter, the debonding of the interface is a very fast phenomenon and that only a small amount of the elastic energy is dissipated to ensure the convergence of the solution. Fig. 6b shows that there is no significant differences when β varies. Therefore, $\Delta\tau\dot{\Delta}/\Delta$ can be neglected in Eq. (23) when compared to the damage variable d . We have to keep in mind that those small differences control the convergence of the finite element analysis.

3.2.2. Estimation of the density of dissipated energy

The previous discussion clearly proves that the viscous term introduced into the bilinear law is the key parameter which guarantees the stability of the solution. However, the physical solution must have a minimal dependence on the choice of the viscous parameter β . To clarify this point, the viscous energy which controls the convergence of the solution should be calculated. The total strain energy released at the interface, or the density of dissipated energy is as follows:

$$\phi = \int_0^{+\infty} t d\Delta \quad (25)$$

By using Eq. (23), the normalized strain energy can be easily expressed as a function of the critical strain energy release rate, G_c and the viscous energy ϕ_v as

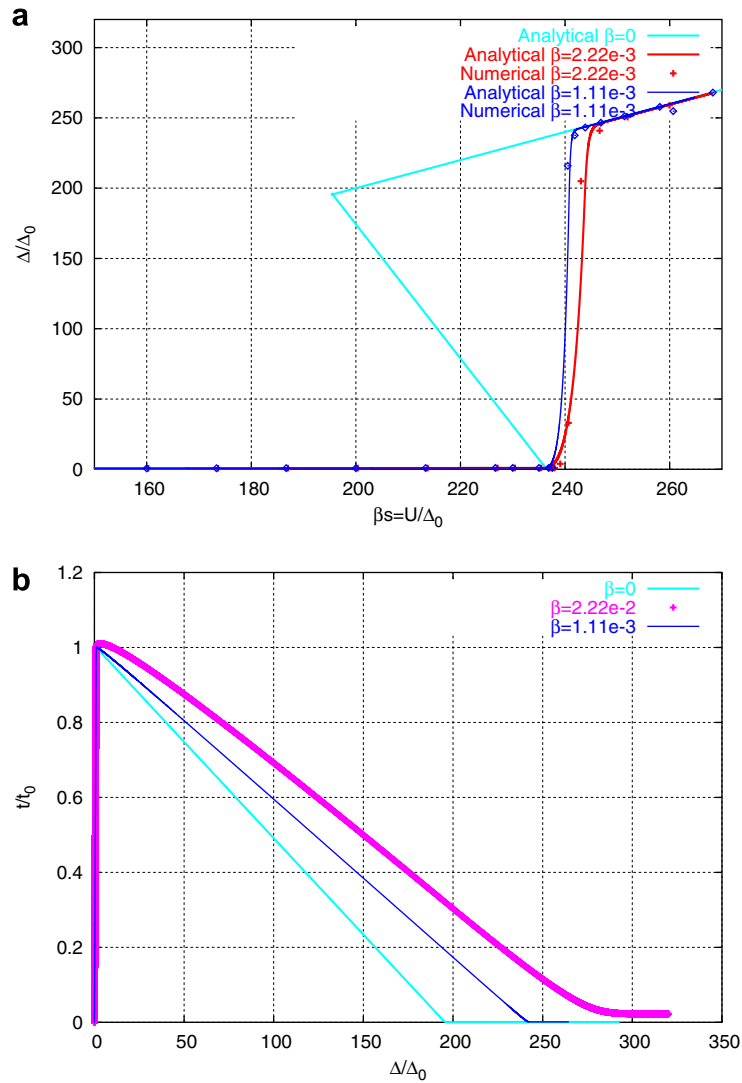


Fig. 5. Response of the cohesive elements for different values of the viscosity: (a) displacement jump versus the imposed displacement and (b) behavior of the cohesive element.

$$\frac{\phi}{\Delta_0 t_0} = \underbrace{\frac{\alpha}{2}}_{G_c/\Delta_0 t_0} + \underbrace{\int_0^{+\infty} \left(\frac{1}{\Delta_0} \frac{d\Delta}{ds} \right)^2 ds}_{\phi_v/\Delta_0 t_0} \quad (26)$$

The stability condition is then derived from the condition of instability given by Eq. (19)

$$\text{The system is stable if : } \frac{\phi_v}{\Delta_0 t_0} \geq \underbrace{\frac{1}{2} \left(\frac{1}{A} + 1 - \alpha \right)}_{\phi_v^{\text{ideal}}/\Delta_0 t_0} \quad (27)$$

where the normalized term $\frac{\phi_v^{\text{ideal}}}{\Delta_0 t_0}$ is called “the ideal energy” and defined as a function of cohesive law parameters and bulk material rigidity. The cohesive strain energy versus the displacement jump (Fig. 7a) is then plotted for different values of the viscosity parameter, β . We show that for small values of β , the cohesive

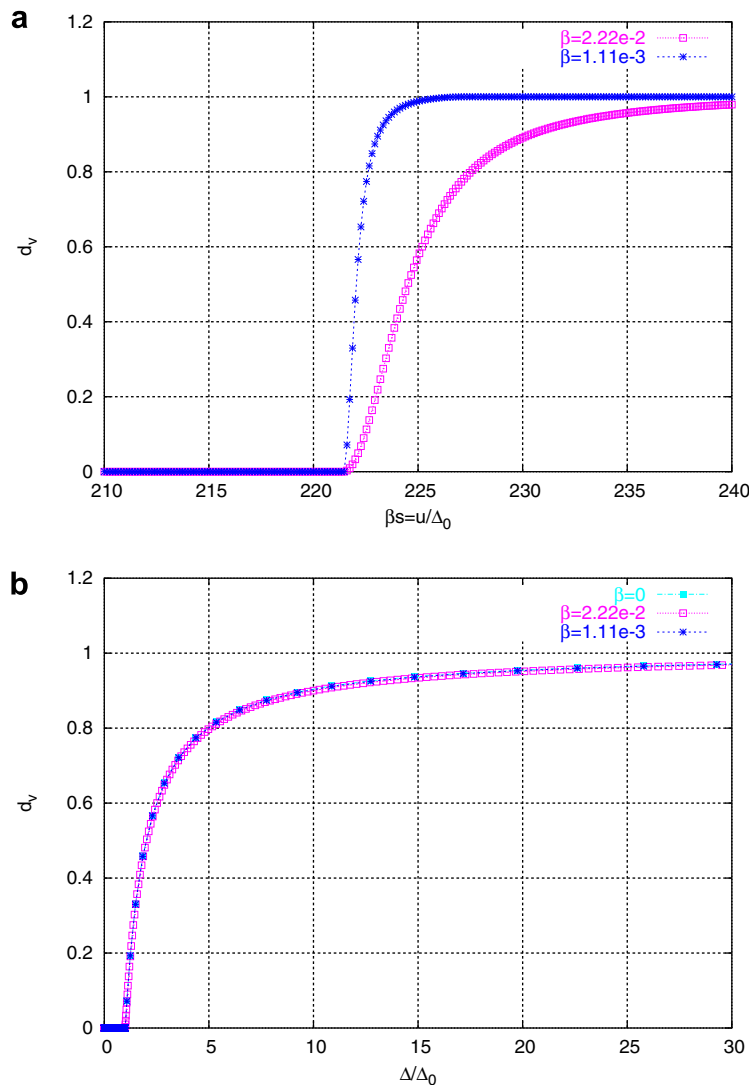


Fig. 6. Evolution of the damage for different values of the viscosity.

energy remains very close to the ideal energy. For larger values of β , the interface energy is slightly higher than the minimum energy but it still ensures the numerical convergence since Eq. (27) is satisfied.

In order to describe the rate of the convergence, the difference between the actual interface energy, which is dissipated by the debonding process, and the ideal energy, defined by Eq. (27), is plotted versus β (Fig. 7b). Finally, the dependence on the viscous parameter β can be approximated by the following law: $\frac{\phi_v - \phi_{ideal}}{A_0 t_0} = \sqrt[n]{\beta/a_0}$, where a_0 and $(n > 1)$ are constants and depend on the material parameters. We notice that the viscous energy is dependent on the imposed displacement rate and is proportional to $\sqrt[n]{V_0}$. We denote that the time-dependent cohesive models presented by Allix et al. [30,31] and Chaboche [17] for rate conditions, the rate dependency appears as quadratic and linear but in our model it is a square root function, which is very satisfactory.

4. Example of a double cantilever beam

The model of the interface has been implemented in ABAQUS Finite Element Implicit Code (ABAQUS v6.5) as a user element subroutine (UEL). The bulk material element selected is a 4-node plane strain

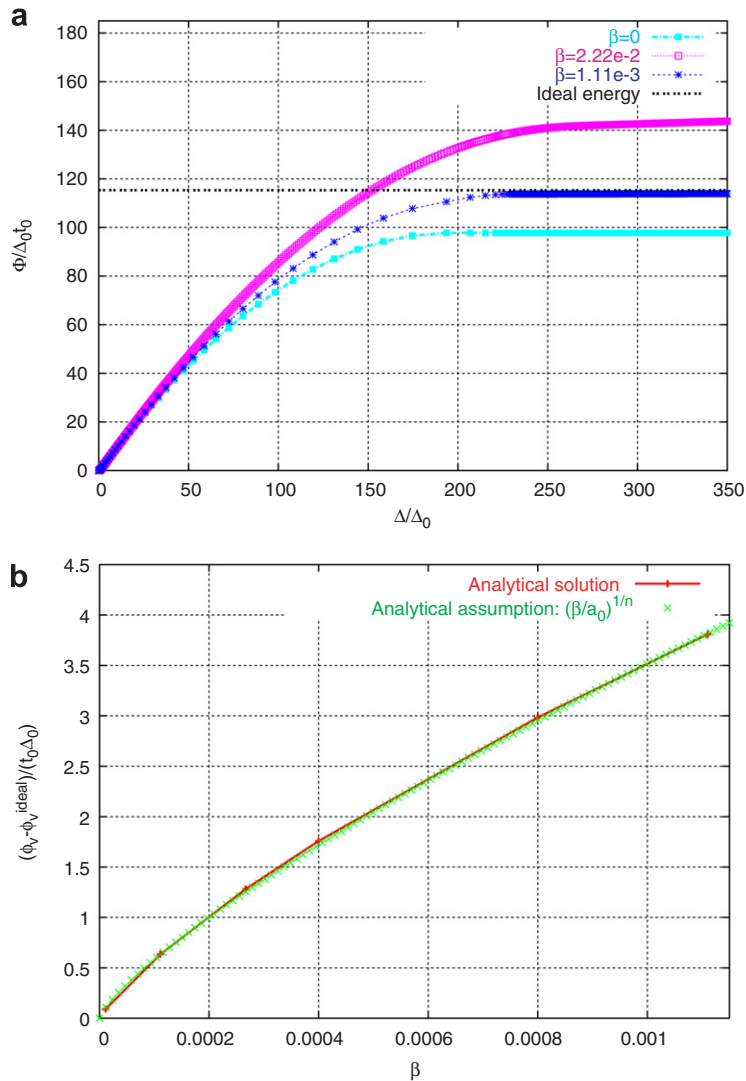


Fig. 7. Sensitivity of the debonding and viscous energies to the viscosity: (a) total debonding energy versus the normalized opening and (b) viscous energy versus the viscosity parameter.

quadrilateral, 4-node interface elements and a Gauss integration are also adopted. In order to evaluate the performances of the regularized model, the numerical predictions are compared with experimental data from double cantilever beam DCB experiments. Therefore, DCB tests on a laminated fibre-reinforced composites glass/vinylester are numerically simulated.

4.1. Mode I fracture toughness test

DCB tests are usually performed to acquire information on the resistance to crack propagation in mode I. This resistance is characterized by the critical energy release rate G_{Ic} . The experiment is conducted according to the ASTM D 5528-94a standard [32]. The specimens are glass taffetas fabrics vinylester DION 9102 matrix reinforced laminated composites. They are 4 mm (10 plies) thick, 21.36 mm wide and 176 mm long. A thin Teflon film is utilized to pre-crack the mid-plane of the laminate. The pre-crack's initial length is a_0 (Fig. 8). A uni-axial testing system mounted on a LLYOD 1 kN max load testing machine is operated in displacement control mode, the crosshead speed is 1 mm/min, and a computer-based data acquisition system is

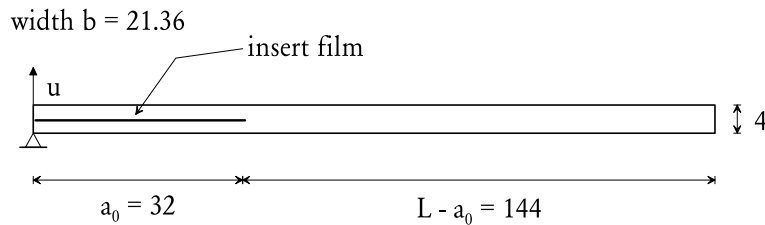


Fig. 8. DCB specimen geometry (dimensions in mm).

used. The crack tip progression is monitored by painting the side of the specimen with a white correction fluid and labeled at 1 mm intervals with a thin blade. During the test, the load–displacement curve is monitored and the crack lengths are continuously measured with a digital camera (resolution 1280×960 pixels, frame rate 7.5 fps) equipped with a $8\times$ zoom.

Three different methods can be utilized to calculate the mode I fracture toughness according to ASTM standards: the compliance calibration method (CC), the Modified Compliance Calibration method (MCC) and the modified beam theory method (MBT). The initiation experimental values of fracture toughness G_{Ii} depends on the specimens because it is strongly linked to the local crack tip conditions [33]. Thus, the average value was $G_{Ii} = 0.5 \text{ kJ/m}^2$. The critical energy release rates are systematically estimated by those three aforementioned methods and is listed in Table 1. The retained mode I fracture toughness was consistent with values reported by other authors for similar materials [34].

The use of woven glass reinforcement for laminated composite induces fiber bridge effects [35]. This is visible in the experimental force–displacement curve. However, it was observed that the delaminations grew at a reasonably constant speed in a stable fashion but, there exists a considerable fluctuation in the G_I of crack extension.

4.2. Finite element model

A 2D 4-node decohesion element is used to simulate DCB tests on woven vinylester glass–fiber-reinforced composite. The model is using 1440 decohesion elements along the length of the specimen. The arms are meshed by a regular quadrilateral plane strain element where, the selected element size is $0.1 \times 0.1 \text{ mm}$, Fig. 9. The boundary conditions are applied by suppressing the degrees of freedom at the bottom end of the left beam. The imposed displacement is applied to the top right ends arms. The critical energy release rate G_{Ic} is taken as input of the cohesive zone model. The maximum interlaminar mode I tension stress t_2^0 is chosen to be $t_2^0 = 50 \text{ MPa}$. This value has been deduced from $(F-U)$ experimental curve. It cannot be completely validated by our present experimental approach because it is difficult to determine the very onset traction of a delamination with accuracy [36]. The penalty parameter should be sufficiently small to limit the specimen's elastic deformation. Several authors already suggested values for the penalty parameter, e.g. Zou et al. [37]. In the present study, the penalty stiffness has been taken as $k_n = 10^6 \text{ N/mm}^3$ in mode I. Table 2 summarizes the different inputs of the model.

Fig. 10 shows the viscosity effect on the response of DCB specimen subjected to pure mode I. One notices that the viscosity parameter significantly affects the response of the model because the cohesive model behavior was modified. Indeed, viscous energy increases with the increase of the viscosity parameter introduced into the model, which induces a rise in resistance to delamination. The choice of this parameter must be regarded as very important. Also, one can say that the decrease in the viscosity parameter generates a significant computational cost. The choice of the “best” viscosity parameter depends on three different points. These are the

Table 1
Critical energy release of the specimens

	Specimens	CC	MCC	MBT	Retained values
G_{Ic} (kJ/m ²)	3	0.987	0.940	0.899	0.900
G_{Ic} (kJ/m ²) [34]	4	–	–	–	1.05

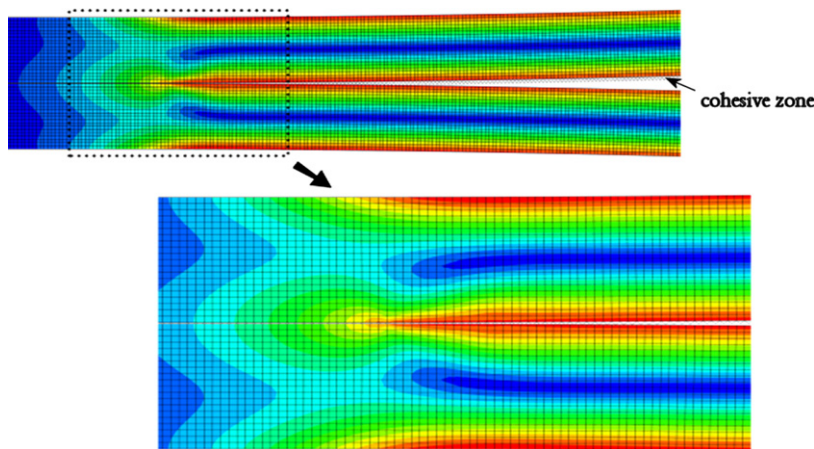


Fig. 9. DCB test finite element meshing.

Table 2
Material data

E_1 (GPa)	E_2 (GPa)	E_3 (GPa)	ν_{12}	ν_{13}	ν_{23}	G_{12} (GPa)	G_{13} (GPa)	G_{23} (GPa)	G_{Ic} (kJ/m ²)
21.6	21.6	9.00	0.18	0.22	0.22	4.66	2.33	2.33	0.9

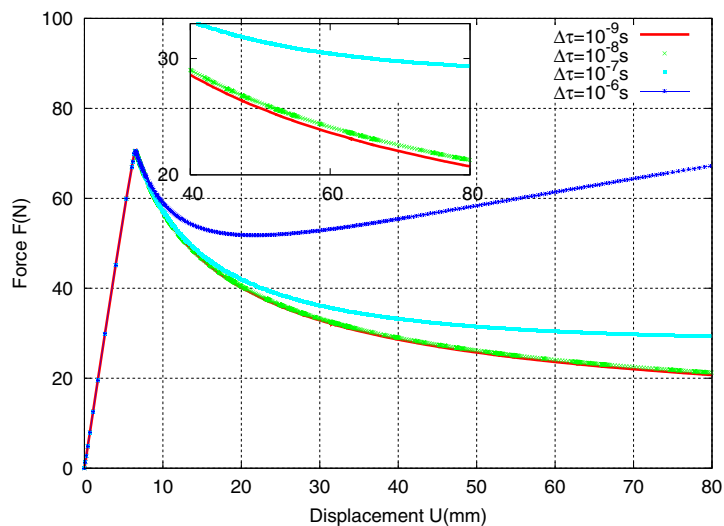


Fig. 10. Force–displacement curves obtained with different viscosity parameter values.

computational cost, its influence on the model behavior and convergence of the numerical solution. By taking account of these different points, the retained value for viscosity is $\Delta\tau = 10^{-8}$ s.

The experimental load–deflection response is first compared to the analytical solution from the linear beam theory in Fig. 11a. Since, the specimens are relatively thin, the transverse shear strain can be neglected. These curves are compared with the numerical finite element simulations obtained by both the constitutive regularized viscous cohesive model law and the bilinear softening, without regularization. In Fig. 11b the crack front extraction is plotted versus the imposed displacement for the experimental, numerical and analytical approaches. These curves are in reasonable agreement with both figures. The specific effect of the viscous regularization on the convergence of the cohesive model is highlighted in Fig. 10a. Indeed, the cohesive model

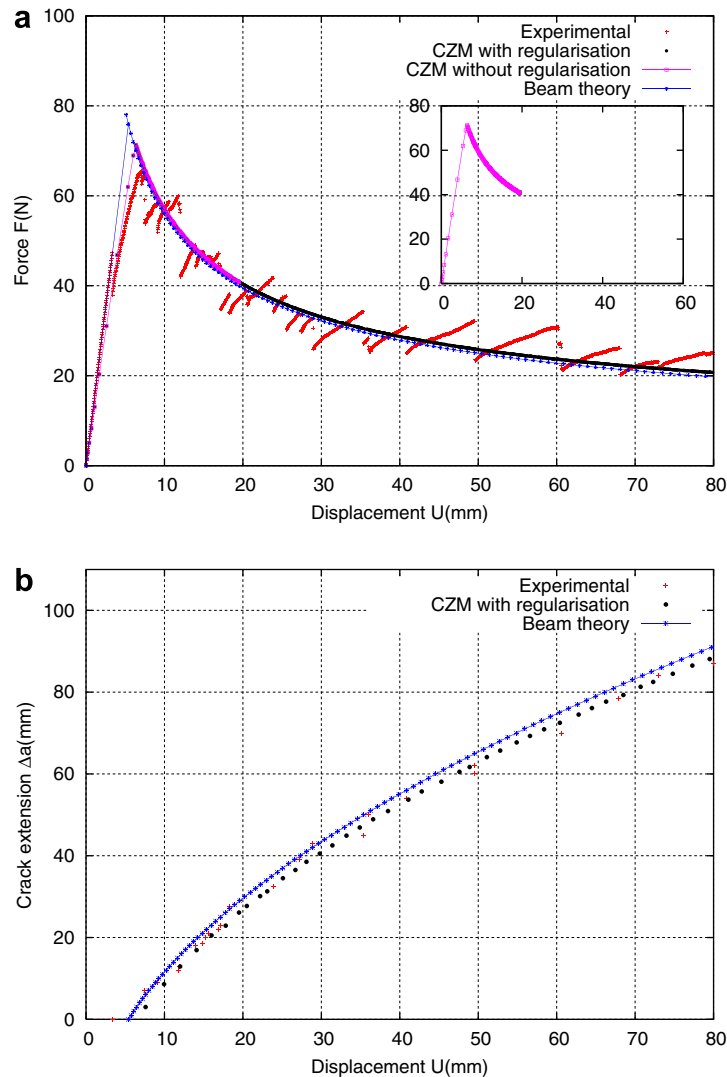


Fig. 11. Numerical-analytical-experimental comparisons: (a) load-displacement curves and (b) curves crack length-displacement.

without regularization induces some problems when the Newton-Raphson algorithm solver cannot overcome to the numerical instabilities.

5. Conclusions

This paper focused on a thermodynamic modeling of the interface softening law. Special attention has been given to the numerical instability problems induced by the possibility of solution jumps in the 1D, 2D or 3D rate independent model. The analysis of a comprehensive 1D case shows that for particular sets of the constitutive cohesive law parameters, the mesh size and the elastic bulk properties, the system can be subject to some instability. The addition of a viscous component to the cohesive model, which permits to meet the thermodynamic admissibility, allows us to overcome this difficulty and to obtain the convergence of the computation. There exist in the literature some interface models with viscous regularization that can eliminate the solution jumps. However, these models induce a great loading rate dependency upon the loading rate. The time-dependent law described in this paper induces minimum loading rate dependency. This is very important for the simulation of the initiation and propagation of the delamination.

We carried out the mode I fracture toughness test with FE simulation to prove the efficiency of the numerical model. From the response of the DCB specimen we have shown a good agreement between experimental data and our proposed model. Indeed, with our regularization technique the initiation stage of damage is not disturbed. Therefore, the initiation criterion of a cohesive model is independent of the viscosity term. In general, the behavior of the model is very satisfactory for the simulation of the initiation and the propagation of delamination.

Finally, one can conclude that the regularization method presented here is a powerful numerical tool to overcome the convergence problem. However, attempts should be made to gain insight of the physical meaning of the introduced viscosity.

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